Efficient Fast Time-Domain Processing Framework for Airborne Bistatic SAR Continuous Imaging Integrated With Data-Driven Motion Compensation

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***Abstract*— Fast factorized back-projection (FFBP) is a classic fast time-domain algorithm (FTDA), which is not limited by the assumption of azimuth-invariant of echo signal and is suitable for the bistatic synthetic aperture radar (BiSAR) process of arbitrary geometric configuration. However, when the conventional FFBP processing is employed for continuous imaging of multiple full-apertures, the processing efficiency will be decreased significantly, and difficulty will be introduced in motion compensation (MOCO) development. The main contributions in this article include the following two aspects: 1) a new FTDA framework based on FFBP implementation is developed for continuous imaging where echo data are divided into several full-aperture data blocks and then processed separately by FFBP implementation to reduce redundant BP operations for achieving high efficiency and 2) an efficient and effective data-driven MOCO methodology is developed based on the new FTDA framework for high focusing quality. In MOCO, because the phase error functions of subimages are estimated in the phase history domain from different local polar coordinate systems, these phase error functions are actually discontinuous in the spatial domain, which will bring significant discontinuity and defocusing into the final image. To address this problem, the correspondence of error functions between the spatial domain and the wavenumber domain is revealed based on which the phase error functions are reconstructed to remove the discontinuity for high focusing quality. Promising results from both simulation and raw data experiments are provided and analyzed to validate the high performance of the proposed algorithm.**

***Index Terms*— Bistatic synthetic aperture radar (BiSAR), continuous imaging, fast factorized back-projection (FFBP), fast time-domain algorithm (FTDA), motion compensation (MOCO).**

Manuscript received December 21, 2020; revised April 30, 2021 and June 18, 2021; accepted July 19, 2021. Date of publication August 2, 2021; date of current version January 17, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 61801204, Grant 62061031, Grant 62065012, and Grant 61665006; in part by the Public Foundation from the Key Laboratory of EMW Information, Fudan University, China, under Grant EMW201901; and in part by the Natural Science Foundation of Jiangxi Province under Grant 20171ACB21018. *(Corresponding author: Song Zhou.)*

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Digital Object Identifier 10.1109/TGRS.2021.3099204

I. Introduction

SYNTHETIC aperture radar (SAR) can observe scenarios in all weather and all day-and-night to generate high-resolution microwave images, which has become an important tool for microwave remote sensing [1], [2]. Bistatic SAR (BiSAR) refers to the SAR system where the transmitter and the receiver are separated on different platforms [3], [4]. Although BiSAR is more complex than conventional monostatic SAR in a geometric configuration, its flexibility allows it to obtain more scattering information from the illuminated area [5]. Moreover, BiSAR has the capability of forward-looking imaging and can be utilized in many applications, such as terminal guidance and attacking, aircraft navigation, and self-landing [6]–[8]. Because of these significant potentials, BiSAR has attracted growing interest in recent years.

Similar to conventional monostatic SAR, the imaging algorithms for BiSAR can be divided into the frequency-domain algorithm (FDA) and time-domain algorithm (TDA) categories. The widely used FDA includes range-Doppler algorithm (RDA), chirp-scaling algorithm (CSA), Omega-K algorithm, and so on, and these algorithms have been well applied in many practical SAR systems. However, the algorithm development will become much more complex when FDAs are extended to many BiSAR applications due to the flexible bistatic configuration and complex signal property [9]–[12]. For example, in general BiSAR, the assumption of azimuth-invariant of the echo signal is no longer available due to the complex application conditions and the poor stability of the radar platforms, which may bring difficulties in the FDA development for BiSAR process [13]–[15].

TDA is the other trend for multistatic SAR applications with arbitrary flying trajectory [16] because it is independent of azimuth-invariant assumption and can avoid complex range-azimuth coupling problems in algorithm development. Thus, TDA has significant advantages in obtaining high focusing performance for the BiSAR process [15]–[17]. Both

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back-projection (BP) and fast factorized BP (FFBP) are famous TDAs, which have high-accuracy imaging applicability in practical systems [18]–[20]. However, compared with the FFBP algorithm, the intensive computation burden of the BP algorithm prevents its advantages from many applications. The FFBP algorithm is a classic fast TDA (FTDA), which can significantly reduce the computation burden and maintains the accuracy and applicability of the BP algorithm [20]–[24]. Moreover, due to the inherent processing architecture, it can be further accelerated by implementing the algorithm on parallel processors for real-time operation [17]. These advantages motivate us to develop more efficient and practical FTDA for BiSAR applications.

In many applications, the SAR sensor is working within long time duration to obtain a wide-swath image in azimuth, which can be referred to as continuous imaging. The typical application is the strip-map mode. However, for continuous imaging, huge redundant BP operations will be inevitably introduced into the FFBP process due to the geometric configuration of continuous imaging and lead the algorithm very inefficient. In this article, a new FTDA framework based on aperture segmentation for continuous imaging is developed. In the new framework, the echo data are divided into several full-aperture data blocks, and each block is processed by the FFBP algorithm separately to form a number of subimages. Then, these subimages are merged in azimuth to obtain the final image. Comparing with the traditional FFBP, the new processing framework can significantly reduce redundant BP operations and achieve high efficiency. Thus, in continuous imaging, the BiSAR imaging algorithm and motion error compensation are developed based on the new processing framework.

In the FFBP process for practical airborne BiSAR applications, the motion error problem needs to be specifically considered [25]. The platform vibration, wind field, and turbulence will bring unknown motion errors, and the impact on the focusing cannot be neglected, especially for the low-cost and small platforms [26], [27]. The motion error can be measured and compensated by the high-accuracy navigation system, such as the inertial measurement unit and global navigation system (IMU/GPS). However, due to the limitation of payload and cost, many small airborne platforms, e.g., UAVs, cannot equip such a high-accuracy navigation system [28]–[30]. Therefore, it is necessary to develop data-driven motion compensation (MOCO) in the SAR process [31]. At present, some MOCO algorithms without navigational data are developed, such as MOCO algorithms in the deep learning-based approach [32], the sparsity-driven approach [33], and RDA [30]. Moreover, there are some ways to combine MOCO with FFBP algorithms for BiSAR applications [34]–[37], which can achieve high performance in focusing quality. However, these MOCO algorithms cannot be directly applied in the new FTDA framework for continuous imaging. Thus, it is still a challenge to develop efficient and effective MOCO under the new framework for continuous imaging.

In the new processing framework for BiSAR continuous imaging, each data block is processed by the FFBP algorithm to achieve a number of subimages. Before merging these subimages to obtain the final image, data-driven MOCO is applied to each subimage to remove the influence of motion error with high efficiency. However, the estimated error functions of subimage are obtained from different polar coordinate systems in the phase history domain so that these estimated error functions are actually discontinuous in the spatial domain. If the error compensated subimages are merged directly, significant discontinuity and defocusing will occur in the final image. Zhang *et al.* [38] proposed a method to reconstruct the phase error function in the Cartesian coordinate for FDA applications. Because these error functions are obtained from a unified Cartesian coordinate system, these error functions can be readily reconstructed to remove their discontinuity. However, when it is based on the new FTDA framework, the error functions are obtained from different polar coordinate systems in the phase history domain, which makes the error reconstruction much more difficult for removing discontinuity, so that the method of [38] cannot achieve high performance. Therefore, it is necessary to consider the problem of phase error discontinuity in the MOCO development under the new FTDA framework.

In this article, a new phase error reconstruction methodology is developed for continuous imaging based on the new framework. First, the analytical spectrum of the subimage in the polar coordinate system is carefully derived by utilizing wavenumber decomposition [39]. Second, based on the analytical spectrum, the correspondence of error functions between the spatial and spectrum domains is revealed. Third, according to the correspondence, the error functions obtained from the image spectrum domain are transformed into the spatial domain for reconstruction. By ensuring that the new error functions are continuous in the spatial domain after reconstruction, the discontinuity and defocusing can be effectively removed from the final SAR image.

This article is organized as follows. In Section II, the computational complexity of conventional FFBP for continuous BiSAR imaging is briefly reviewed, based on which the new processing framework is developed for continuous imaging. In Section III, the subimage spectrum of BiSAR with motion error is derived, and data-driven MOCO for one single full-aperture is developed. Then, in Section IV, the correspondence of error component between spectrum and spatial domains is revealed, based on which a new phase error reconstruction methodology is developed to remove the discontinuity and defocusing from the final image. Simulation and raw data experiments are provided and analyzed in Section V. Finally, the conclusion is drawn in Section VI.

II. Efficient Processing Framework for Continuous Imaging

In this section, the computational complexity of FFBP is carefully investigated, and a new FTDA framework is developed for continuous imaging with high efficiency. For continuous BiSAR imaging, it generally contains several full-apertures for achieving a wide observing swath in azimuth. First, we briefly review the concept of full-aperture and the complexity of BP and FFBP algorithms, as shown in Fig. 1(a). Assume that the range samples on the formed image are M,

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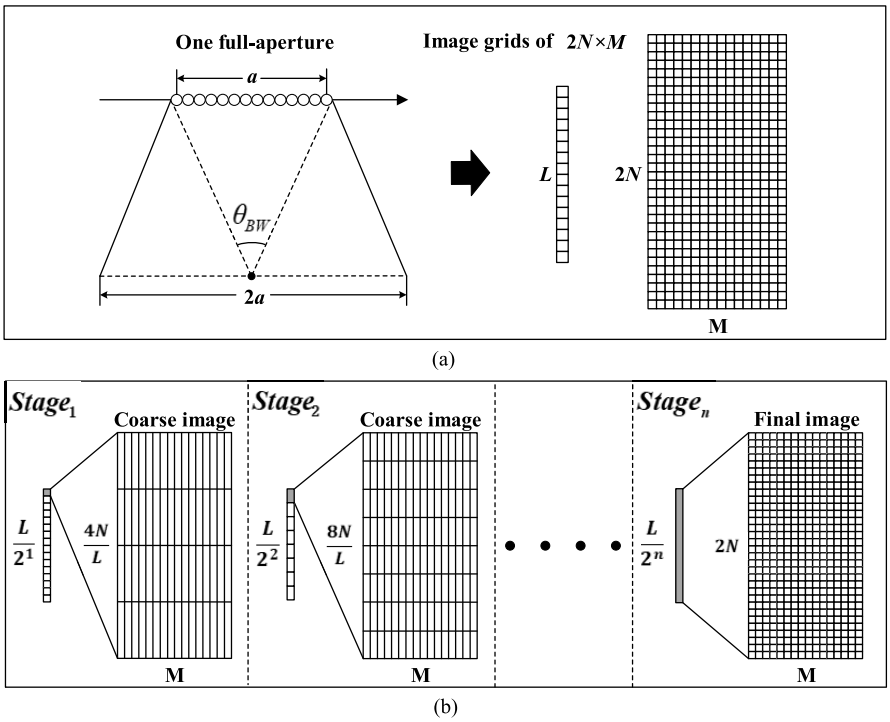


Fig. 1. Concept of full-aperture and implementation of traditional FFBP. (a) Concept of full-aperture. (b) Computational complexity of traditional FFBP for one full-aperture.

and the synthetic array length of one single full-aperture is a, which is mainly limited by the synthetic beamwidth of θ_{BW} containing L azimuth samples. In Fig. 1(a), for one single full-aperture with L samples, the observing range in azimuth is about $2a$. To facilitate the following derivations, we assume that the full-resolution in azimuth is a/N, and it can be calculated that the total samples in azimuth on the formed image is $2N$; then, the total number of BP operations for one full-aperture imaging is given by

$$L \times 2N \times M = 2MNL. \tag{1}$$

The conventional FFBP algorithm adopts the idea of multistage merging strategy in the polar coordinate system [20]. Let $L = 2^n$, and the FFBP process for one single full-aperture contains n recursions. The range resolution of the image scenario is assumed to be constant in each recursion. In the first processing stage, assume that the full-aperture is divided into $L/2$ subapertures, and each subaperture contains 2 azimuth samples. Then, echoes from every subaperture are utilized to form coarse images. The azimuth resolution corresponding to the subaperture is degraded by $L/2$ times compared with that of full-aperture [20], and the number of BP operations in the first processing stage is given by

$$\mathrm{Stage}_1 : 2 \times \frac{2a}{\left(\frac{a}{N} \times \frac{L}{2}\right)} \times M \times \frac{L}{2} = 4MN \tag{2}$$

where $(a/N \times L/2)$ is the sampling interval in azimuth on one formed coarse image, and M is the range samples, which remains the same as the implementation in the BP algorithm. In the second processing stage, every two coarse images are merged together so that $L/4$ coarse images can be obtained; then, the azimuth resolution is improved by 2 times compared with that of the previous stage. The number of BP operations in the second processing stage is given by

$$\mathrm{Stage}_2 : 2 \times \frac{2a}{\left(\frac{a}{N} \times \frac{L}{4}\right)} \times M \times \frac{L}{4} = 4MN \tag{3}$$

where $(a/N \times L/4)$ is the sampling interval in azimuth on the formed coarse image, which is half of that on the formed coarse image in the first stage, and M is the range samples, which remains the same as the previous stage. In the third processing stage, every two coarse images are merged, and $L/8$ coarse images can be obtained; then, the azimuth resolution is improved by two times again compared with that of the previous stage. The number of BP operations in the third processing stage is given by

$$\mathrm{Stage}_3 : 2 \times \frac{2a}{\left(\frac{a}{N} \times \frac{L}{8}\right)} \times M \times \frac{L}{8} = 4MN. \tag{4}$$

As the length of subaperture increasing recursively, the azimuth resolution of the coarse image is improved accordingly until it reaches the highest resolution corresponding to one full-aperture, as shown in Fig. 1(b). According to the above recursive process, the number of BP operations in the

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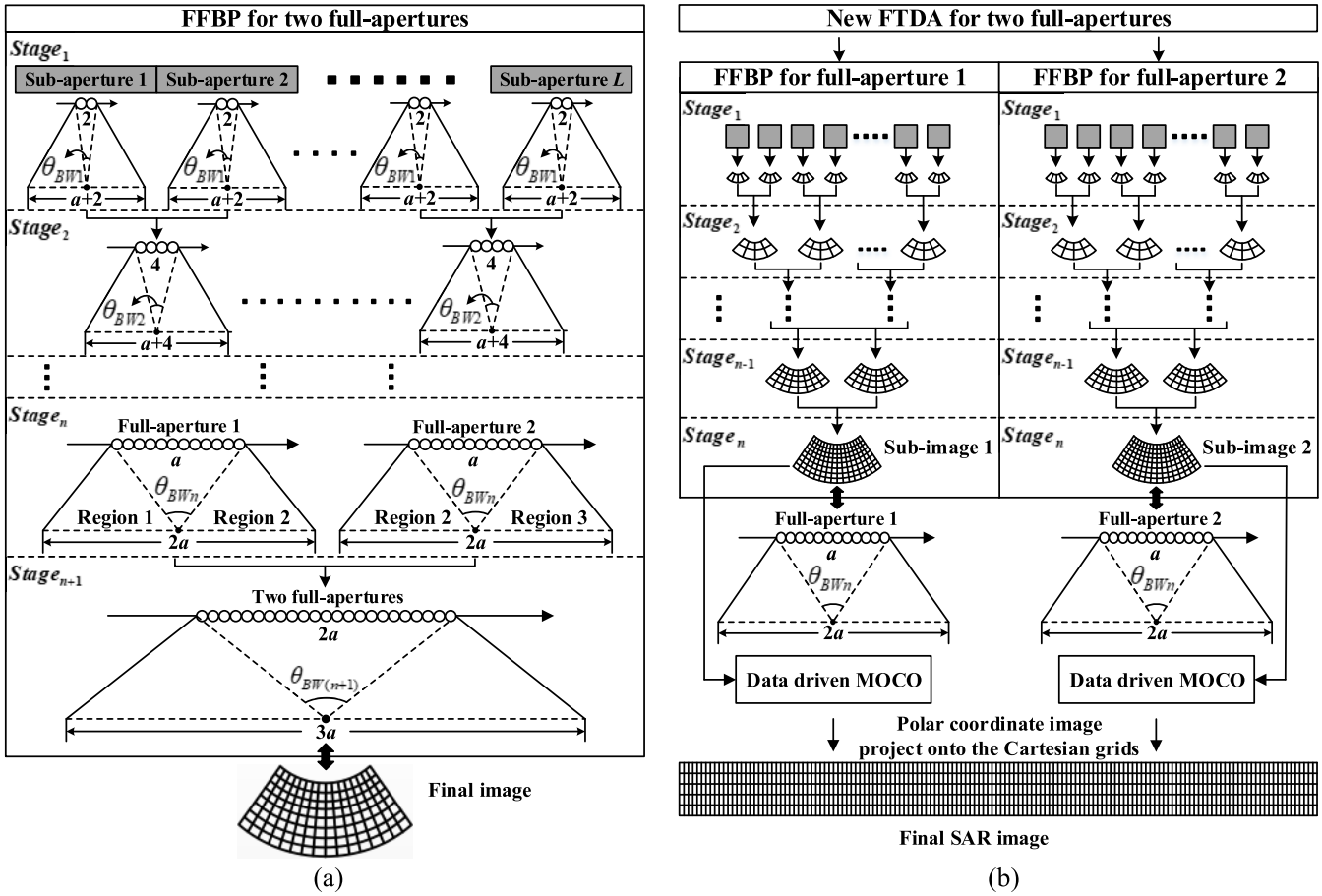


Fig. 2. Different FTDAs for continuous imaging. (a) Conventional FTDA for two full-apertures. (b) New FTDA framework for two full-apertures.

remaining processing stages is given by

$$\begin{cases} \text{Stage}_4 : 2 \times \dfrac{2a}{\left(\dfrac{a}{N} \times \dfrac{L}{16}\right)} \times M \times \dfrac{L}{16} = 4MN \\ \quad \vdots \\ \text{Stage}_n : 2 \times \dfrac{2a}{\left(\dfrac{a}{N} \times \dfrac{L}{2^n}\right)} \times M \times \dfrac{L}{2^n} = 4MN. \end{cases} \tag{5}$$

After n stages of image merging processing, the final full-resolution image of one single full-aperture is obtained. Equations (2)–(5) show that each stage has the same number of BP operations. The total number of BP operations is given as

$$n \times (4MN) = 4MN \log_2 L. \tag{6}$$

Comparing (6) with (1), the computational complexity of the conventional FFBP algorithm is improved by $L/(2\log_2 L)$ times.

Second, we consider continuous imaging of echoes from two full-apertures, which is based on the conventional FFBP framework, and the process for two full-apertures is represented in Fig. 2(a). The azimuth length of two full-apertures becomes $2a$, and the illuminated range in azimuth becomes $3a$, as shown in Fig. 2(a). If traditional FFBP is directly applied to the two full-apertures echoes, the azimuth samples of two full-apertures are $2^{n+1} = 2L$, and the process contains $(n+1)$ stages. Then, it can be known from (5) that the number of BP operations in each stage from stage 1 to stage n is given by

$$\begin{cases} \text{Stage}_1 : 2 \times \dfrac{3a}{\left(\dfrac{a}{N} \times \dfrac{L}{2}\right)} \times M \times \dfrac{2L}{2} = 12MN \\ \quad \vdots \\ \text{Stage}_n : 2 \times \dfrac{3a}{\left(\dfrac{a}{N} \times \dfrac{L}{2^n}\right)} \times M \times \dfrac{2L}{2^n} = 12MN. \end{cases} \tag{7}$$

In the nth stage, the azimuth resolution of the image has reached the finest value of N, and the resolution will no longer be improved in the $(n+1)$th stage. Therefore, the number of BP operations of the $(n+1)$th stage can be given as

$$\text{Stage}_{n+1} : 2 \times \frac{3a}{\left(\dfrac{a}{N} \times \dfrac{L}{2^n}\right)} \times M \times \frac{2L}{2^{n+1}} = 6MN. \tag{8}$$

Therefore, with (7) and (8), the number of BP operations of the conventional FFBP for two full-apertures imaging is given as

$$n \times (12MN) + 6MN = (12MN) \log_2 L + 6MN. \tag{9}$$

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Comparing (6) and (9), it can be noted that the length of the aperture is doubled; however, the number of BP operations is nearly tripled. This is due to the fact that lots of redundant BP operations are introduced into the FFBP implementation, which will significantly increase the computational burden in the process. As shown in Fig. 2(a), the illuminated range of full-aperture 1 only contains region 1 and 2, while the illuminated range of full-aperture 2 only contains regions 2 and 3. However, in the conventional FFBP implementation, the echoes from full-aperture 1 are utilized for projecting in region 3; meanwhile, the echoes from full-aperture 2 are utilized for projecting in region 1. In fact, echoes from full-aperture 1 will not contribute to the imaging of region 3, and echoes from full-aperture 2 will not contribute to the imaging of region 1. However, in this strategy, both the imaging of region 3 of full-aperture 1 and the imaging of region 1 of full-aperture 2 are utilized in the final fusion stage. This is the reason why the redundant BP operations are introduced in continuous imaging for full-aperture 1 and full-aperture 2. In the FFBP development for continuous imaging, we need to reduce the number of redundant BP operations in the implementation for achieving high efficiency.

To reduce redundant BP operations, a new processing framework is proposed, as shown in Fig. 2(b), where the echo data are divided into two separated full-aperture data blocks. In the new processing framework, the echoes from each full-aperture are processed by FFBP separately, which contains n recursive stages. In the $(n+1)$th stage, the subimages are merged onto unified Cartesian grids to obtain the final SAR image. Based on new processing framework and according to (6), the number of BP operations for two full-apertures is given as

$$2 \times n \times (4MN) = 8MN \log_2 L. \tag{10}$$

Comparing (9) and (10), the efficiency of the new framework is improved evidently. Because the echoes from each full-aperture are processed separately, the echoes from full-aperture 1 will not be utilized for projecting in region 3; meanwhile, the echoes from full-aperture 2 will not be utilized for projecting in region 1. Therefore, the redundant BP operations can be significantly reduced. Here, we just consider the case of continuous imaging, including only two full-apertures. In the case of several full-apertures, the efficiency improvement will become much more significant. Considering continuous imaging of echoes including Q full-apertures, the total azimuth samples are QL. Let μ_Q denote the number of total BP operations of traditional FFBP implementation with Q full-apertures and λ_Q denote the number of total BP operations of the new framework with Q full-apertures; then, the improvement coefficient of η_Q is denoted and given by

$$\eta_Q = \frac{\mu_Q}{\lambda_Q}. \tag{11}$$

The larger η_Q is, the more significant improvement it indicates. Assume that $L = M = a$, where $L = 1024$. For $Q = 2$, 4, 6, and 8, the corresponding η_Q for continuous imaging is shown in Fig. 3. From Fig. 3, it can be seen that the improvement becomes much more evident as the number of full-apertures increases, which shows the superior performance of the new processing framework. In the following sections, the MOCO development for airborne BiSAR will be also based on the new framework for high processing efficiency.

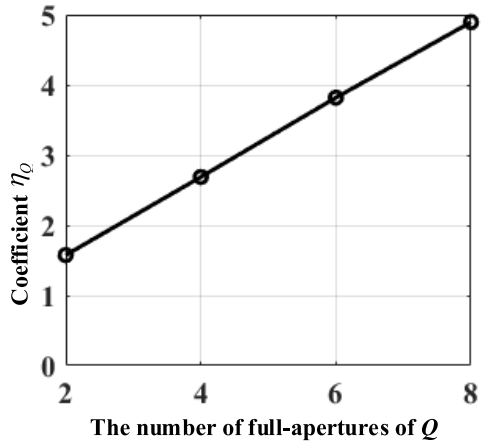


Fig. 3. Improvement of the new FTDA framework for continuous imaging.

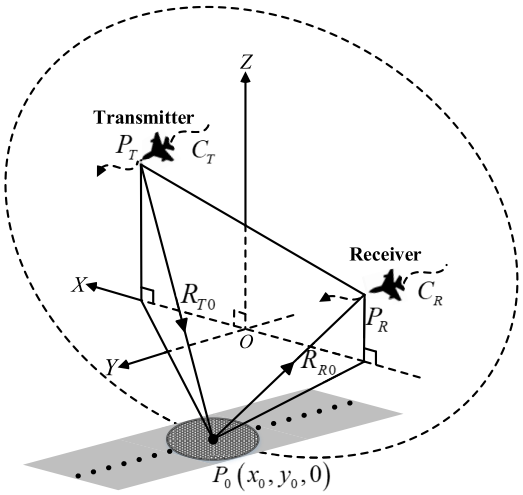


Fig. 4. BiSAR geometry in the OPE coordinate system.

III. MOCO for One Single Full-Aperture Imaging

For the airborne BiSAR process, the motion error problem needs to be specifically considered. As shown in Fig. 2(b), the data-driven MOCO is also integrated into the new processing framework for high efficiency and high focusing quality. However, the estimated error function of each subimage is obtained from different local polar coordinate systems. If the compensated subimages are merged directly, significant discontinuity and defocusing will inevitably occur in the final image. In this section, we first develop data-driven MOCO for airborne BiSAR imaging of one single full-aperture, which is an extension of the algorithm of [31]. The solution to the problem of discontinuity and defocusing for obtaining the final image in continuous imaging will be discussed in Section IV.

A. BiSAR Signal Model of One Single Full-Aperture

Fig. 4 shows the BiSAR geometric model. The transmitter is mounted on an aircraft moving along curve C_T, while the

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receiver is mounted on another platform moving along curve C_R. The continuously illuminated area is shown in Fig. 4, and P_0 denotes an arbitrary point scatterer in the scenario. Let t denote the azimuth time variable. At t, the antenna phase center (APC) of the transmitter is at $P_T(x_T, y_T, z_T)$, while the APC of the receiver is at $P_R(x_R, y_R, z_R)$, where x_T, y_T, z_T, x_R, y_R, and z_R can be regarded as functions with variable of t. Similarly, x_T, y_T, z_T, x_R, y_R, and z_R can be regarded as functions with variable of t, and it has

$$\begin{cases} (x_T, y_T, z_T) = \left[f_{x_T}(t), f_{y_T}(t), f_{z_T}(t)\right] \\ (x_R, y_R, z_R) = \left[f_{x_R}(t), f_{y_R}(t), f_{z_R}(t)\right]. \end{cases} \tag{12}$$

In real BiSAR geometry, we need 3-D coordinates of grids for range calculation. To simplify the derivation of the algorithm, we assume that the point scatterers in the illumination area are on the ground plane with zero altitudes. If the digital elevation model (DEM) information is available in practical applications, the DEM information needs to be carefully incorporated into the FFBP process [40]. R_T denotes the instantaneous range of transmitter to the point scatterer, and R_R denotes the instantaneous range of the receiver to the point scatterer. At t, the echoes from P_0 can be given as

$$S(t, K) = \alpha \exp\left(-j\vec{R}_{T0} \cdot \vec{K}_T - j\vec{R}_{R0} \cdot \vec{K}_R\right) \tag{13}$$

where "·" denotes inner product operation, and α denotes a constant reflectivity coefficient. $\vec{K}_T$ is the wavenumber vector of the transmitted signal in the direction from P_T to P_0, and $\vec{R}_{T0}$ is the range vector in the same direction as $\vec{K}_T$. $\vec{K}_R$ is the wavenumber vector of the echo signal in the direction from P_R to P_0, and $\vec{R}_{R0}$ is the range vector in the same direction as $\vec{K}_R$. In (13), $\vec{K}_T$ and $\vec{K}_R$ have the same magnitude that is denoted by K, and the magnitudes of $\vec{R}_{T0}$ and $\vec{R}_{R0}$ are denoted by R_{T0} and R_{R0}, given as

$$R_{T0} = \sqrt{(x_0 - x_T)^2 + (y_0 - y_T)^2 + (0 - z_T)^2} \tag{14}$$

and

$$R_{R0} = \sqrt{(x_0 - x_R)^2 + (y_0 - y_R)^2 + (0 - z_R)^2} \tag{15}$$

respectively.

In direct BP implementation, the echoes are projected onto Cartesian grids of $(x_g, y_g, 0)$ to form an SAR image, and the BP image can be given by the following integral [16]:

$$i\left(x_g, y_g, 0\right) = \int s(t, R_\Sigma)|_{R_\Sigma = R_T + R_R} dt \tag{16}$$

where

$$s(t, R_\Sigma) = \int S(t, K) \exp(jR_\Sigma K) dK. \tag{17}$$

In (16) and (17), R_T is the range from P_T to image grid of $P(x, y, 0)$, which can be denoted as the range vector $\vec{R}_T$, and R_R is the range from P_R to the image grid of $P(x, y, 0)$, which can be denoted as the range vector $\vec{R}_R$. R_Σ is the sum of R_T and R_R. For R_T and R_R, it has

$$R_T = \sqrt{(x - x_T)^2 + (y - y_T)^2 + (0 - z_T)^2} \tag{18}$$

and

$$R_R = \sqrt{(x - x_R)^2 + (y - y_R)^2 + (0 - z_R)^2}. \tag{19}$$

With (13) and (17)–(19), (16) can be rewritten as

$$i\left(x_g, y_g, 0\right) = \int\int \alpha \exp\left(-j\vec{R}_{T0} \cdot \vec{K}_T - j\vec{R}_{R0} \cdot \vec{K}_R\right) \times \exp\left(+j\vec{R}_T \cdot \vec{K}_T + j\vec{R}_R \cdot \vec{K}_R\right) dK dt. \tag{20}$$

Let $\vec{R}_0$ denote the range vector from the origin O to P_0 and $\vec{R}$ the range vector from O to P so that $(\vec{R}_0 - \vec{R})$ denotes the range vector from P to P_0, and (20) can be rewritten as [40]

$$i\left(x_g, y_g, 0\right) = \int\int \alpha \exp\left[-j\left(\vec{R}_0 - \vec{R}\right) \cdot \left(\vec{K}_T + \vec{K}_R\right)\right] dK dt. \tag{21}$$

Let $\vec{K}_\Sigma = \vec{K}_T + \vec{K}_R$ denote the combined wavenumber vector from both transmitter and receiver, and(21) can be reformed as

$$i\left(x_g, y_g, 0\right) = \int\int \alpha \exp\left[-j\left(\vec{R}_0 - \vec{R}\right) \cdot \vec{K}_\Sigma\right] dK dt \tag{22}$$

where $(\vec{R}_0 - \vec{R})$ in the first term indicates the location of P_0 focused on the image plane after direct BP process. Then, the real case with motion error is considered. APC of the transmitter with motion error is assumed at $P'_T(x'_T, y'_T, z'_T)$, and the APC of the receiver with motion error is assumed at $P'_R(x'_R, y'_R, z'_R)$. The real trajectories of the transmitter and receivers are denoted by the dashed curves C_T and C_R, respectively. Let ΔR_T denote the motion error of the transmitter and ΔR_R denote the motion error of the receiver. Both ΔR_T and ΔR_R can be also regarded as functions of azimuth time t. For simplicity, let $i(x_g, y_g, 0) = i(x_g, y_g)$; then, based on the reference [40] and (13)–(22), it obtains

$$i\left(x_g, y_g\right) \approx \int\int \alpha \exp\left[-j\Delta\vec{R}_T \cdot \vec{K}_T - j\Delta\vec{R}_R \cdot \vec{K}_R\right] \times \exp\left[-j\left(\vec{R}_0 - \vec{R}\right) \cdot \vec{K}_\Sigma\right] dK dt \tag{23}$$

where the first term of (23) represents the phase error introduced by the unknown motion error.

Then, we consider the influence of motion error in the FFBP process. In the process, both FFBP and MOCO are developed based on an orthogonal elliptical polar (OEP) system for high efficiency. The advantages and details of OEP can be found in [40]. Considering one single full-aperture from t_1 to t_2, the ideal trajectories of the transmitter and receivers are denoted by the curves of C_1 and C_2, respectively. Assume that, at center azimuth time $t = t_c$, the APCs of transmitter and receiver are at F_1 and F_2, respectively. When motion error is introduced, the real trajectories of the transmitter and receivers are denoted by the dashed curves of C'_1 and C'_2, respectively, and the APCs of transmitter and receiver with motion error are at F'_1 and F'_2, respectively. As shown in Fig. 5, the OEP coordinate system at the center azimuth time of t_c is described by the dashed ellipse with the focuses of F_1 and F_2. In the OEP system, the equivalent azimuth phase center (EAPC) is defined at $O_\perp$, and $\angle F_1 P_0 F_2$ is equally divided by $\vec{O_\perp P_0}$. $\theta_{\perp 0}$

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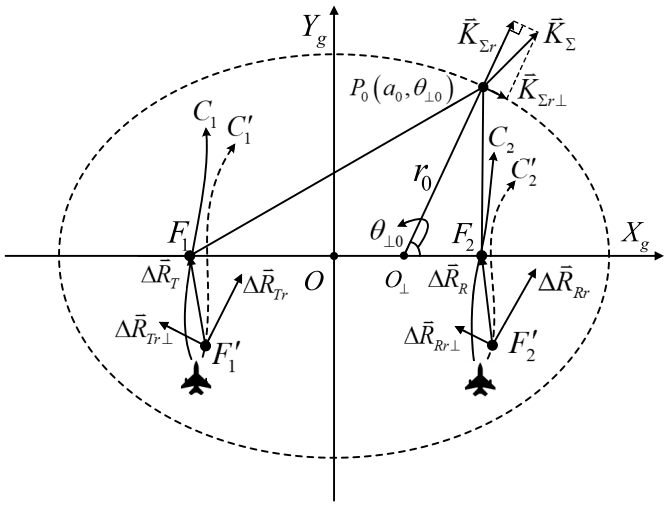


Fig. 5. Wavenumber vectors decomposed in the OEP coordinate system.

is the angle between $\vec{O_\perp P_0}$ and $\vec{F_1 F_2}$; thus, the coordinate of P_0 is represented as $i(a_0, \theta_{\perp 0})$, where a_0 is defined by

$$a_0 = \frac{1}{2}\left(|\vec{F_1 P_0}| + |\vec{F_2 P_0}|\right) \tag{24}$$

where $|\cdot|$ denotes the magnitude calculation for the vector. To derive the image spectrum in the OEP system, a pair of wavenumber vectors is introduced, denoted as $\vec{K}_r$ and $\vec{K}_{r\perp}$. Then, the wavenumber vectors of signal in (23) are decomposed along $\vec{K}_r$ and $\vec{K}_{r\perp}$ directions, which can be given as

$$\begin{cases} \vec{K}_\Sigma = \vec{K}_{\Sigma r\perp} + \vec{K}_{\Sigma r} \\ \vec{K}_T = \vec{K}_{\mathrm{Tr}\perp} + \vec{K}_{\mathrm{Tr}} \\ \vec{K}_R = \vec{K}_{\mathrm{Rr}\perp} + \vec{K}_{\mathrm{Rr}} \end{cases} \tag{25}$$

where the wavenumbers $K_{\Sigma r\perp}$, $K_{\mathrm{Tr}\perp}$, and $K_{\mathrm{Rr}\perp}$ have the same direction as $K_{r\perp}$, and $K_{r\perp}$ can be regarded as the intersection of $K_{\Sigma r\perp}$, $K_{\mathrm{Tr}\perp}$, and $K_{\mathrm{Rr}\perp}$. Similarly, the wavenumbers $K_{\Sigma r}$, K_{Tr}, and K_{Rr} have the same direction as K_r, and K_r can be regarded as the intersection of $K_{\Sigma r}$, K_{Tr}, and K_{Rr}. Then, it has

$$\begin{cases} \vec{K}_{r\perp} = \vec{K}_{\Sigma r\perp} \cap \vec{K}_{\mathrm{Tr}\perp} \cap \vec{K}_{\mathrm{Rr}\perp} \\ \vec{K}_r = \vec{K}_{\Sigma r} \cap \vec{K}_{\mathrm{Tr}} \cap \vec{K}_{\mathrm{Rr}}. \end{cases} \tag{26}$$

For an arbitrary grid in OEP system, BiSAR image represented in OEP is given by $i(a, \theta_\perp)$, and (23) can be transformed into the following integral [40]:

$$\begin{aligned} i(a,\theta_\perp) \approx \int\int & \alpha \exp(-j\Delta R_{r\perp} K_{r\perp} - j\Delta R_r K_a \cos\beta_0) \\ & \times \exp(-j\theta_{\perp 0} r_0 K_{r\perp} - j a_0 K_a) \\ & \times \exp(j\theta_\perp r_0 K_{r\perp} + j a K_a) \cos\beta_0 dK_{r\perp} dK_a \end{aligned} \tag{27}$$

where

$$\begin{cases} \beta_0 = \dfrac{1}{2}\angle F_1 P_0 F_2 \\ r_0 = \left|\vec{O_\perp P_0}\right| \\ K_a = K_r / \cos\beta_0. \end{cases} \tag{28}$$

In (27), K_r and $K_{r\perp}$ are the magnitudes of $\vec{K}_{r\perp}$ and $\vec{K}_r$. $\Delta R_{r\perp}$ and ΔR_r are the magnitudes of $\Delta\vec{R}_{r\perp}$ and $\Delta\vec{R}_r$, where $\Delta\vec{R}_{r\perp}$ denotes the motion errors of both transmitter and receiver projected along $\vec{K}_{r\perp}$ direction, while $\Delta\vec{R}_r$ denotes the motion errors of both transmitter and receiver projected along $\vec{K}_r$ direction, given as

$$\begin{cases} \Delta\vec{R}_r = \Delta\vec{R}_{\mathrm{Tr}} + \Delta\vec{R}_{\mathrm{Rr}} \\ \Delta\vec{R}_{r\perp} = \Delta\vec{R}_{\mathrm{Tr}\perp} + \Delta\vec{R}_{\mathrm{Rr}\perp}. \end{cases} \tag{29}$$

In (29), $\Delta\vec{R}_{\mathrm{Tr}}$, $\Delta\vec{R}_{\mathrm{Rr}}$, $\Delta\vec{R}_{\mathrm{Tr}\perp}$, and $\Delta\vec{R}_{\mathrm{Rr}\perp}$ are derived from the decomposition of $\Delta\vec{R}_T$ and $\Delta\vec{R}_R$, as shown in Fig. 5. In addition, K_a is the wavenumber variable with respect to a, while $K_{r\perp}$ is the wavenumber variable with respect to $\theta_\perp$. In (28), the relationship between wavenumber K_a and K_r is revealed, and the details of them can be found in [40]. Then, the signal spectrum $I(K_a, K_{r\perp})$ can be obtained from (27), and it has

$$\begin{aligned} I(K_a, K_{r\perp}) \approx \alpha & \exp(-j\Delta R_{r\perp} K_{r\perp} - j\Delta R_r K_a \cos\beta_0) \\ & \times \exp(-j\theta_{\perp 0} r_0 K_{r\perp} - j a_0 K_a)\cos\beta_0. \end{aligned} \tag{30}$$

Equation (30) reveals the presentation of motion error in the image spectrum domain. In Section III-B, data-driven MOCO will be developed based on (30).

B. Data-Driven MOCO for One Single Full-Aperture

Let $\varphi_e(K_a, K_{r_\perp})$ be the phase error function of (30), given as

$$\varphi_e(K_a, K_{r\perp}) \approx -\Delta R_{r\perp} K_{r\perp} - \Delta R_r K_a \cos\beta_0 \tag{31}$$

where $\Delta R_{r\perp}$ and ΔR_r can be regarded as functions of K_r and $K_{r\perp}$, given as

$$\begin{cases} \Delta R_r = \Delta R_r(K_r, K_{r\perp}) \\ \Delta R_{r\perp} = \Delta R_{r\perp}(K_r, K_{r\perp}). \end{cases} \tag{32}$$

The relationship between azimuth phase error (APE) and nonsystematic range cell migration (NsRCM) is revealed by applying first-order Taylor expansion with respect to $K_a = K_{a0}$ [31], [41], [42], where K_{a0} is denoted as the central wavenumber, and (31) can be approximated as

$$\varphi_e(K_a, K_{r\perp}) \approx \varphi_0(K_{a0}, K_{r\perp}) + \varphi_1(K_{a0}, K_{r\perp})\Delta K_a \tag{33}$$

where

$$\varphi_1(K_{a0}, K_{r\perp}) = \frac{1}{K_{a0}}\varphi_0(K_{a0}, K_{r\perp}) + \frac{\partial\varphi_0(K_{a0}, K_{r\perp})}{\partial K_{r\perp}}\frac{K_{r\perp}}{K_{a0}} \tag{34}$$

where $\Delta K_a = K_a - K_{a0}$. The first item of (33) is APE, which can be estimated directly by the classical autofocusing methods [43]–[45]. Then, the estimated phase error is denoted by $\varphi_0(K_{r\perp})$, which is a function with respect to $K_{r\perp}$ in the phase history domain. The second item of (33) is the NsRCM, which can be derived from estimated $\varphi_0(K_{r\perp})$. Then, the relationship between APE and NsRCM is given in (34). From (34), it is seen that the first item of NsRCM has a

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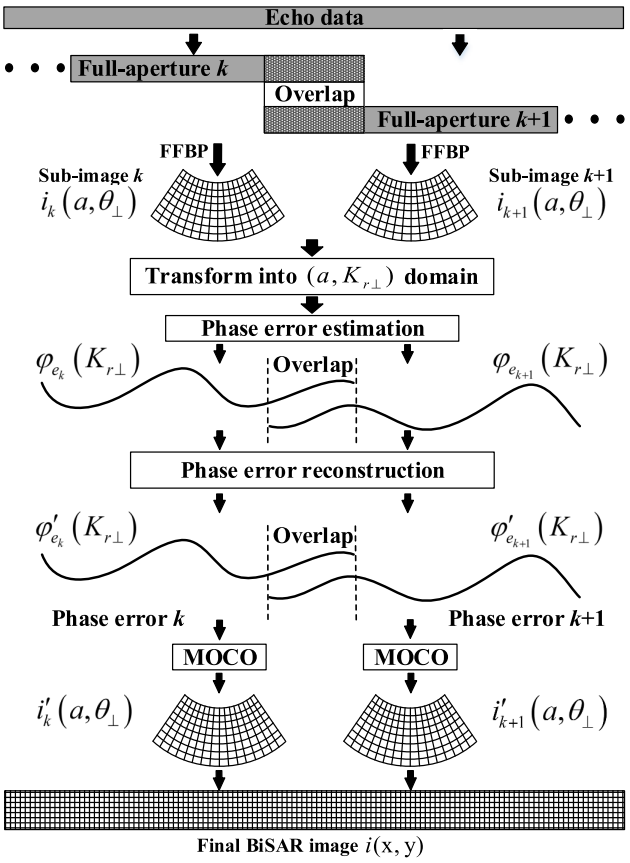


Fig. 6. MOCO for continuous imaging under the new FTDA framework.

linear relationship with APE, and the second item of NsRCM has a linear relationship with the derivative of APE. With estimated $\varphi_0(K_{r\perp})$, correction for both APE and NsRCM can be implemented according to (33) and (34), and the well-focused image can be obtained.

So far, we have developed data-driven MOCO for BiSAR imaging of one single full-aperture. In Section IV, efficient and effective MOCO methodology will be developed for continuous imaging, which contains several full-aperture data processes.

IV. Phase Error Reconstruction and MOCO for Continuous Imaging

In Section III, we discussed data-driven MOCO for one single full-aperture imaging. In continuous imaging, it generally contains several full-aperutures to obtain a wide observing swath in azimuth. As shown in Fig. 2(b), data-driven MOCO is implemented after n recursive processing stages. If the compensated subimages from different full-apetures are merged directly, significant discontinuity and defocusing will occur in the final image. This is because the error functions estimated from different full-apertures are spatially discontinuous. Thus, it is necessary to reconstruct these error functions to remove discontinuity and defocusing from the final image for high focusing quality.

In this section, a new MOCO methodology is developed, as shown in Fig. 6. From it, we can see that echo data are divided into several full-aperture data blocks, and the adjacent full-apertures overlap with each other. Then, subimages from several full-apertures are obtained by the FFBP process. After that, the error function is estimated from the phase history domain by autofocusing techniques, e.g., phase gradient autofocus (PGA), map-drift autofocus (MDA), and minimum entropy autofocus (MEA) [43]–[45]. The overlapped part of the segments is used for phase error reconstruction and connection between adjacent full-apertures [38]. By utilizing phase error estimation, the obtained error function of all subimages in the polar grid is from different polar coordinate systems. Thus, it is necessary to find the relationship between the phase error functions of the adjacent subimages for reconstruction and ensure that they are continuous in the spatial domain.

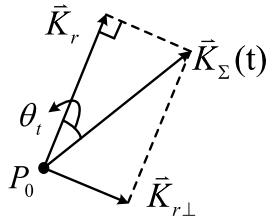


Fig. 7. Relationship among θ_t, K_Σ, and K_r.

As shown in Fig. 6, the subimages in the $(a, \theta_\perp)$ domain are transformed into the $(a, K_{r_\perp})$ domain for phase error estimation. For subimage k and $k+1$, the estimation errors are expressed as $\varphi_{e_k}(K_{r_\perp})$ and $\varphi_{e_{k+1}}(K_{r_\perp})$, respectively, in the phase history domain. From Fig. 6, it can be noted that $\varphi_{e_k}(K_{r_\perp})$ and $\varphi_{e_{k+1}}(K_{r_\perp})$ are obtained from the phase history domain in different polar coordinate systems. If these error functions become continuous in the spatial domain, the final image can be accurately focused without discontinuity and defocusing. Therefore, we consider transforming $\varphi_{e_k}(K_{r_\perp})$ and $\varphi_{e_{k+1}}(K_{r_\perp})$ into the spatial domain to reconstruct these error functions. Due to the complex BiSAR geometry, the relationship among variable of t, the wavenumber of K_r, and $K_{r\perp}$ is very complex so that it is difficult to transform $\varphi_{e_k}(K_{r_\perp})$ and $\varphi_{e_{k+1}}(K_{r_\perp})$ into functions with variable of t. Thus, a new intermediate variable of θ_t is introduced, which denotes the angle between K_Σ and K_r at t, as shown in Fig. 7. The relationship between θ_t and wavenumber variables can be expressed as

$$\theta_t = \arctan\left(\frac{K_{r\perp}}{K_r}\right). \tag{35}$$

Then, with (28), $K_{r\perp}$ can be given by

$$K_{r\perp} = K_{a0}\cos\beta_0\tan\theta_t \tag{36}$$

where K_{a0} is the center of K_a. By this way, the correspondence of phase error function between the phase history domain and the spatial domain of θ_t is obtained, and the error functions can be transformed into the spatial domain of θ_t accordingly.

Then, we consider the reconstruction of the phase error function for two adjacent subimages of k and $k+1$, and their geometry and coordinate systems are shown in Fig. 8. According to (36), the phase errors estimated from subimage

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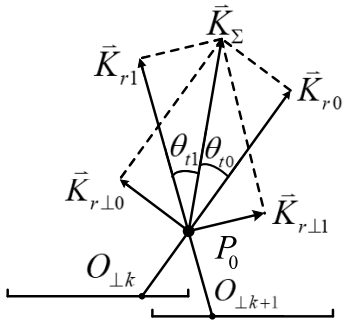


Fig. 8. Spatial correspondence between subimage k and subimage $k+1$ in the adjacent coordinate system.

k and $k+1$ are given by

$$\begin{cases} \varphi_{e_k}\left(K_{r_{\perp 0}}\right) = \varphi_{e_k}(\theta_{t0}) \\ \varphi_{e_{k+1}}\left(K_{r_{\perp 1}}\right) = \varphi_{e_{k+1}}(\theta_{t1}) \end{cases} \tag{37}$$

where θ_{t0} and θ_{t1} are the spatial-domain variables of subimages k and $k+1$. θ_{t0} and θ_{t1}, as well as corresponding wavenumber vectors of subimage k and $k+1$, are also described in Fig. 8. Let $K_{r\perp 0}$ be an arbitrary wavenumber of subimage k. It can be considered that $K_{r\perp 0}$ is a component of K_{Σ}, denoted as $K_{\Sigma}\sin(\theta_{t0})$. Meanwhile, K_{Σ} can also be decomposed on the OEP coordinate system of subimage $k+1$, and it has $K_{r\perp 1} = K_{\Sigma}\sin(\theta_{t1})$, where $K_{r\perp 1}$ is the wavenumber in subimage $k+1$. From Fig. 8, it can be noted that the variables of θ_{t0}, θ_{t1}, $K_{r\perp 0}$, and $K_{r\perp 1}$ have one-to-one correspondence. Therefore, the phase error function of adjacent subimages can be reconstructed according to the one-to-one correspondence of θ_{t0} and θ_{t1}. In addition, the functional relationship between θ_{t0} and θ_{t1} can be also established by utilizing BiSAR geometries corresponding to subimages, as shown in Fig. 9. In Fig. 9, the receiver and the transmitter move along the trajectory of C_R and C_T, respectively. Let t_0 be the azimuth time of subimage k and t_1 be the azimuth time of subimage $k+1$. Then, the BiSAR geometry corresponding to subimage k is described by the plane of $P_{T0}P_0P_{R0}$, and the BiSAR geometry corresponding to subimage $k+1$ is described by the plane of $P_{T1}P_0P_{R1}$, as shown in Fig. 9. From Fig. 9, it can be noted that r_0 is the magnitude of range vector $\vec{O_{\perp k}P_0}$ with respect to the wavenumber vector $\vec{K}_{r0}$ of subimage k, while r_1 is the magnitude of range vector $\vec{O_{\perp k+1}P_0}$ with respect to the wavenumber vector $\vec{K}_{r1}$ of subimage $k+1$. Then, the relationship between r_0 and r_1 can be given as

$$\theta_{ts} = \arccos\left(\frac{{r_0}^2 + {r_1}^2 - \Delta O_{\perp}{}^2}{2r_0r_1}\right) \tag{38}$$

where $\Delta O\perp$ denotes the range from $O_{\perp k}$ to $O_{\perp k+1}$, and θ_{ts} denotes the angle between $\vec{O_{\perp k}P_0}$ and $\vec{O_{\perp k+1}P_0}$, which can be also regarded as the angle between wavenumber vector $\vec{k}_{r0}$ and $\vec{k}_{r1}$. Then, it has

$$\theta_{ts} = \theta_{t0} + \theta_{t1}. \tag{39}$$

With (39), the correspondence between θ_{t0} and θ_{t1} in the spatial domain is established. By substituting (38) into (39),

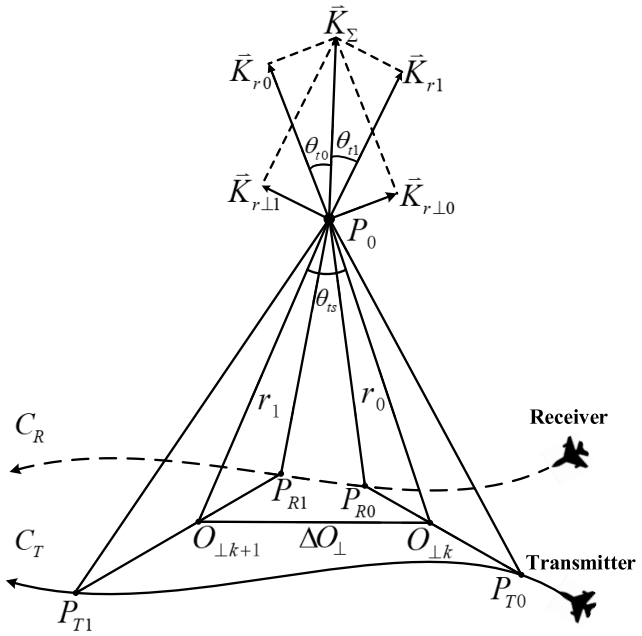


Fig. 9. Geometric relationship between θ_{t0} and θ_{t1}.

θ_{t1} can be obtained by

$$\theta_{t1} = \arccos\left(\frac{{r_0}^2 + {r_1}^2 - \Delta O_{\perp}{}^2}{2r_0r_1}\right) - \theta_{t0}. \tag{40}$$

For simplicity, let function of $H(.)$ denote the relation between θ_{t1} and θ_{t0}, and (40) can be rewritten as

$$\theta_{t1} = H(\theta_{t0}). \tag{41}$$

Then, by substituting (41) into (37), the relationship between adjacent phase errors in the spatial domain of θ_{t0} and the phase history domain is established, given by

$$\begin{cases} \varphi_{e_k}\left(K_{r_{\perp 0}}\right) = \varphi_{e_k}(\theta_{t0}) \\ \varphi_{e_{k+1}}\left(K_{r_{\perp 1}}\right) = \varphi_{e_{k+1}}[H(\theta_{t0})]. \end{cases} \tag{42}$$

In (42), the phase error functions of two adjacent subimages in the same spatial domain of θ_{t_0} are obtained. Subsequently, reconstruction will be carried out in the spatial domain of θ_{t_0} to ensure that $\varphi_{e_k}(K_{r_{\perp 0}})$ and $\varphi_{e_{k+1}}(K_{r_{\perp 1}})$ are continuous in the spatial domain of θ_{t_0}.

The process of phase error reconstruction is shown in Fig. 10. First, according to (36), the adjacent phase errors $\varphi_{e_k}(K_{r_{\perp 0}})$ and $\varphi_{e_{k+1}}(K_{r_{\perp 1}})$ are transformed into the respective spatial domain, denoted as $\varphi_{e_k}(\theta_{t0})$ and $\varphi_{e_{k+1}}(\theta_{t1})$. Second, the phase error in the spatial domain of θ_{t1} is transformed into the spatial domain of θ_{t0}; then, the phase error corresponding to $\varphi_{e_{k+1}}(\theta_{t1})$ in the referential spatial domain of θ_{t0} is obtained and denoted as $\varphi_{e_{k+1}}[H(\theta_{t0})]$. Finally, the overlapped segment of adjacent phase error is utilized for extracting the coherent information to combine $\varphi_{e_k}(\theta_{t0})$ and $\varphi_{e_{k+1}}[H(\theta_{t0})]$ in the spatial domain of θ_{t0} and make sure that $\varphi_{e_k}(\theta_{t0})$ and $\varphi_{e_{k+1}}[H(\theta_{t0})]$ are continuous in the spatial domain of θ_{t0}. Then, the reconstructed phase error $\varphi_{e_{\Sigma}}(\theta_{t0})$ is transformed into the phase history domain. The whole processing diagram is according to Fig. 6 in which the step of phase error reconstruction is

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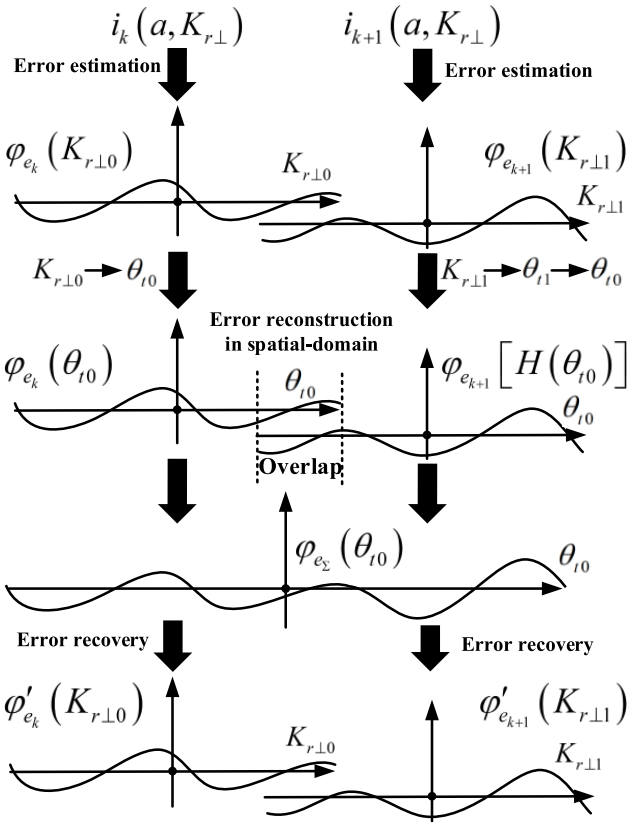


Fig. 10. Process of phase error reconstruction.

TABLE I
SIMULATION POINTS' LOCATION

	Points location		
	P_1	P_2	P_3
	(950,-50,0) m	(1000,-50,0) m	(1050,-50,0) m
	P_4	P_5	P_6
	(950,0,0) m	(1000,0,0) m	(1050,0,0) m
	P_7	P_8	P_9
	(950,50,0) m	(1000,50,0) m	(1050,50,0) m

shown in Fig. 10. As the reconstructed error functions become continuous in the spatial domain, MOCO is subsequently implemented to the subimages, and the discontinuity and defocusing can be removed from the final image for achieving high focusing quality.

V. SIMULATIONS AND RAW DATA EXPERIMENTS

In this section, both simulations and raw data experiments for BiSAR continuous imaging are presented and analyzed to evaluate the performance of the proposed algorithm.

A. Simulation Experiments

In the simulation experiments, the geometries of simulation are in accordance with Fig. 4, and the locations of points 1–9 are given in Table I. The BiSAR parameters are listed in Table II. The bistatic APCs are given by $(x_T, y_T, z_T) = P_T + V_T t + (1/2)a_T t^2$ and $(x_R, y_R, z_R) = P_R + V_R t + (1/2)a_R t^2$, respectively. In the simulation, the time duration

TABLE II
BISAR PARAMETERS IN SIMULATION

Mode	Strip-map
Bandwidth	200 MHz
PRF	1000 Hz
Center Location	(1000,0,0) m
Transmitter Location P_T	(-600,0,1200) m
Receiver Location P_R	(600,0,300) m
Velocity Vector V_T	(0,50,0) m/s
Acceleration Vector a_T	(-0.2,0.3,-0.1) m/s^2
Velocity Vector V_R	(-5,20,0) m/s
Acceleration Vector a_R	(0.1,-0.2 0.3) m/s^2

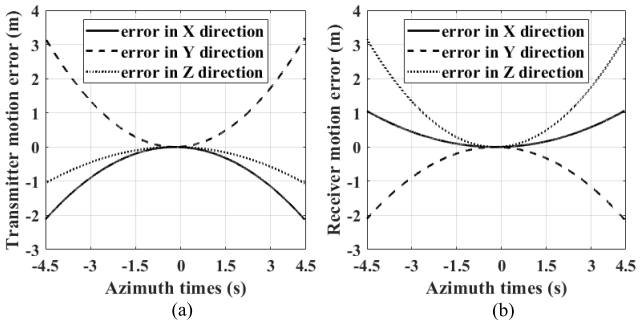


Fig. 11. Motion errors of the transmitter and the receiver in 3-D. (a) Error along the X-, Y-, and Z-directions of the transmitter. (b) Errors along the X-, Y-, and Z-directions of the receiver.

of the synthetic aperture is 9 s, and the assumed motion errors of each platform along the X-, Y-, and Z-directions are shown in Fig. 11. The algorithms for both simulation and raw data experiments are programed on a MATLAB platform and windows 10 system running on a PC with i7-7700 CPU and 16-GB RAM without parallel processing. The experimental results will be analyzed to show the effectiveness of the methodology of phase error reconstruction for removing the image discontinuities.

Based on the processing procedures of the new FTDA framework, the echo data are divided into two full-aperture data blocks, where the number of full-aperture data blocks is determined by the proportional relationship between the beamwidth of the synthetic aperture and the beamwidth of the full-aperture. The time duration of every full-aperture is 5 s, and the adjacent full-apertures with 20% overlapping and then subimages from two full-apertures are obtained by the FFBP process. The total time consumption by the new FTDA framework for BiSAR continuous imaging is 1.6 min, which is much faster than 2.7 min obtained by the traditional FFBP algorithm. In addition, Pu *et al.* [30] proposed the RDA with high efficiency, and the total time consumption by Pu *et al.* [30] for BiSAR continuous imaging is 1.41 min, which is a little faster than that of the proposed algorithm and shows the high efficiency of the proposed algorithm.

The phase error is estimated by utilizing weighted PGA (WPGA) [47]. Because the estimated phase errors are obtained from the phase history domain in different polar coordinate systems separately, phase error reconstruction is implemented in the MOCO to ensure that the error functions are continuous

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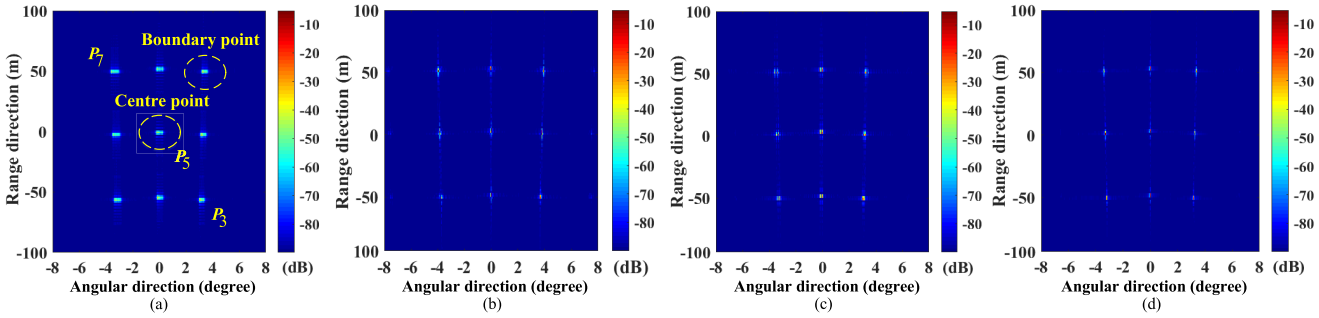


Fig. 12. Multipoint simulation results based on the new continuous imaging framework. (a) Simulation results without MOCO. (b) MOCO results obtained by the traditional method of [46]. (c) MOCO results obtained by the error reconstruction method of [38]. (d) MOCO results obtained by the proposed error reconstruction method.

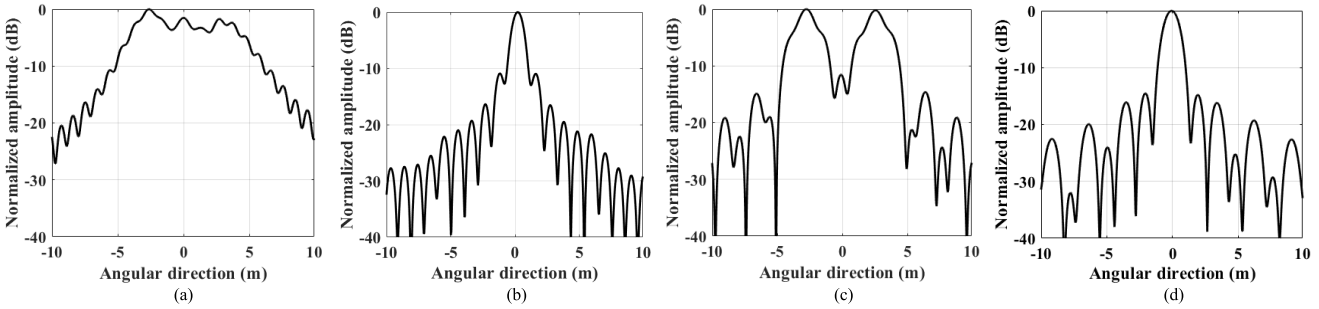


Fig. 13. Azimuth profiles of center point from simulation. (a) Center point without MOCO. (b) Center point obtained by the traditional method of Wu *et al.* [46]. (c) Center point obtained by the error reconstruction method of Zhang *et al.* [38]. (d) Center point obtained by the proposed error reconstruction method.

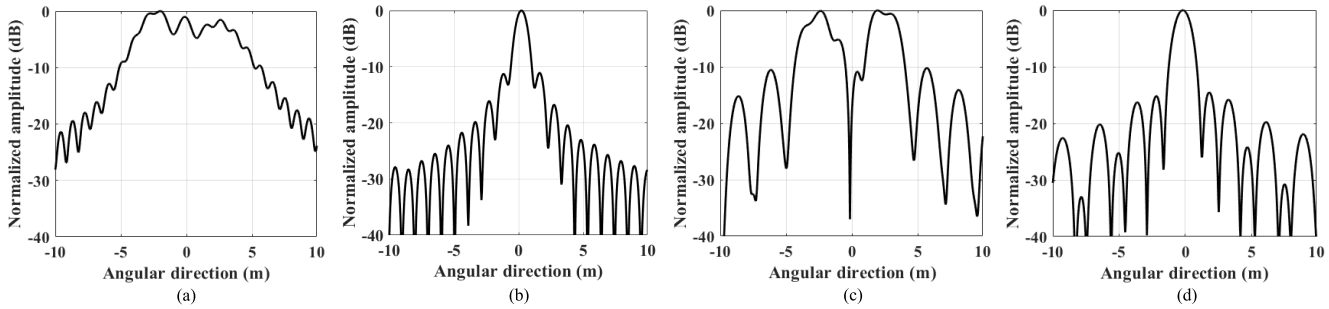


Fig. 14. Azimuth profiles of boundary point from simulation. (a) Boundary point without any MOCO. (b) Boundary point obtained by the traditional method of Wu *et al.* [46]. (c) Boundary point obtained by the error reconstruction method of Zhang *et al.* [38]. (d) Boundary point obtained by the proposed error reconstruction method.

in the spatial domain so that the discontinuity and defocusing can be removed from the final image. Results of BiSAR continuous imaging from the multipoint simulation are based on the new processing framework and different MOCO algorithms, as shown in Fig. 12. Fig. 12(a) shows the BiSAR image without MOCO, and it can be noted that serious defocusing occurs. Fig. 12(b) presents MOCO results by utilizing the traditional method of [46]. From Fig. 12(b), it is seen that the serious defocusing has been removed obviously. Fig. 12(c) presents MOCO results by utilizing the error reconstruction method of [38]. From Fig. 12(c), it is seen that merged point results contain "double mainlobes" as the phase error functions are still discontinuous in the spatial domain, and discontinuity and defocusing are introduced to the final image. By utilizing the proposed error reconstruction method, MOCO results are shown in Fig. 12(d). By comparing Fig. 12(c) with (d), it can be seen that the proposed method can achieve higher focusing quality, and the discontinuity and defocusing are removed from the final image.

To further evaluate the performance of the proposed method, results of both center and boundary points are particularly analyzed, which are highlighted in Fig. 12(a). The azimuth profiles of the center point are shown in Fig. 13, and the azimuth profiles of the boundary point are shown in Fig. 14. Figs. 13(a) and 14(a) present the azimuth profiles of the center point and the boundary point without MOCO. Figs. 13(b) and 14(b)

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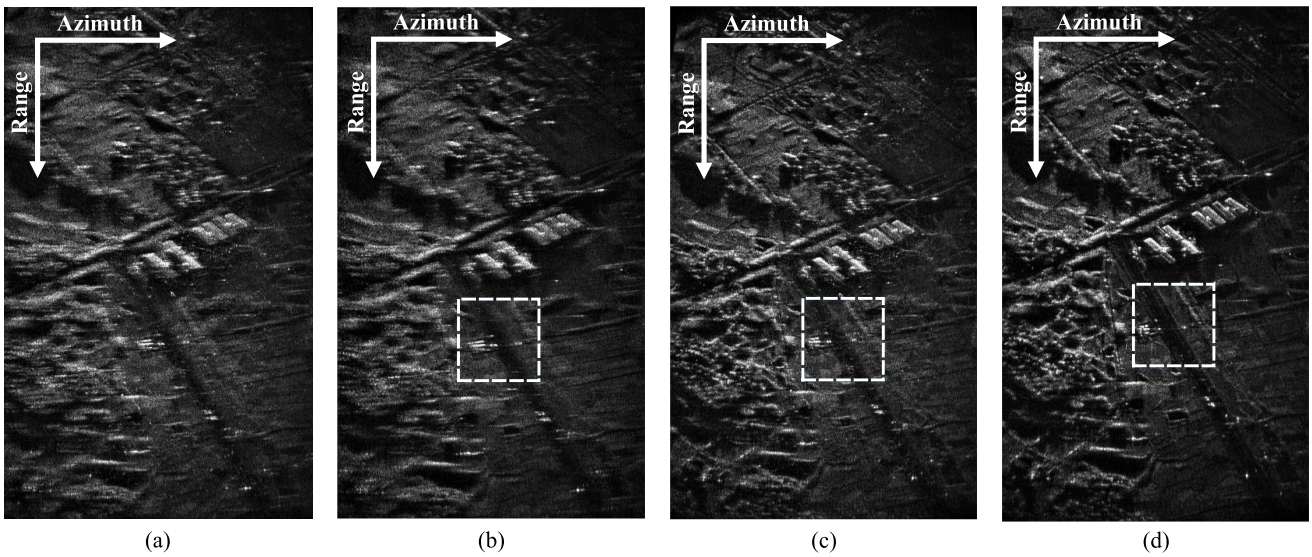


Fig. 15. Final BiSAR images based on the new continuous imaging framework. (a) Results without MOCO. (b) MOCO results obtained by the traditional method of Wu *et al.* [46]. (c) MOCO results obtained by the error reconstruction method of Zhang *et al.* [38]. (d) MOCO results obtained by the proposed error reconstruction method.

present the azimuth profiles of the center point and the boundary point obtained by the conventional method of [46]. From Figs. 13(b) and 14(b), it is seen that the focusing quality improved obviously, but minor defocusing still remains due to the residual phase error, which is caused by the inaccurate error estimation. Especially, if the traditional method is applied to continuous imaging with long time duration, accurate estimation of phase error will become much more difficult. Therefore, it is necessary to develop the new FTDA framework for continuous imaging with a long time duration. Figs. 13(c) and 14(c) present the azimuth profiles of the center point and the boundary point obtained by the error reconstruction method of [38]. From Figs. 13(c) and 14(c), it is seen that significant discontinuity is contained in the image, and "double mainlobes" exist in the azimuth profiles, which shows that the error reconstruction method of [38] cannot achieve high performance in the new FTDA framework. Then, Figs. 13(d) and 14(d) present the azimuth profiles of the center point and the boundary point obtained by the proposed error reconstruction method. Comparing Fig. 14(c) with (d), it can be seen that the discontinuity is removed, and the "double mainlobes" disappear in the final image. Meanwhile, by comparing Figs. 12(d) with 14(b), it can be seen that the proposed method is superior to the conventional method of [46], and the well-focused results validate the superior performance of the proposed method.

B. Raw Data Experiments of Airbone BiSAR

In this section, the performance of the proposed algorithm is further evaluated by experiments with raw BiSAR data. In the experiment, the transmitter is mounted on an aircraft with a velocity of about 65 m/s at an altitude of about 1200 m. The system is operating at the Ku-band with a bandwidth of 80 MHz. Moreover, the transmitter is working on the side-looking mode with a small squint-angle. Meanwhile, the receiver is fixed on a hill near the illuminating scenario. The altitude of the fixed receiver is about 300 m, the baseline between the transmitter and the receiver is about 1500 m, and the range of the scenario center is about 7700 m. The image area is about 2.7 km $\times$ 1.1 km (range direction $\times$ azimuth direction).

In the processing of raw BiSAR data, the azimuth time duration of the aperture is about 17 s. The echo data are segmented into three full-aperture data blocks with 50% overlapping, and the time duration of every full-aperture is about 8 s. Because the INS/GPS information is not accurate enough, the phase errors are extracted from the received echoes for MOCO instead of utilizing navigation data. The raw data processing is running on the same platform mentioned in the simulation section. By utilizing the traditional method of [46], the total time consumption is about 21.1 min, where the total time consumption for the traditional FFBP without MOCO algorithm is about 20.6 min. Then, by utilizing the proposed method, the total time consumption is about 10.2 min, where the total time consumption for continuous imaging framework without data-driven MOCO algorithm is about 9.5 min, which shows higher efficiency of the new FTDA framework for BiSAR continuous imaging. The final BiSAR images based on the new continuous imaging framework are presented in Fig. 15. Fig. 15(a) shows the result without any MOCO showing that the image is totally defocused by the unexpected motion error, and the entropy of the image is 12.930. By utilizing the conventional method of [46], the MOCO result is shown in Fig. 15(b). From Fig. 15(b), it is seen that serious defocusing remains in the final image, which is caused by the inaccurate error estimation in the continuous imaging for a long time duration, and the entropy of the image is 12.634. Therefore, it is necessary to apply the new FTDA framework for continuous imaging. After the error reconstruction process, the whole error functions of three continuous full-apertures are

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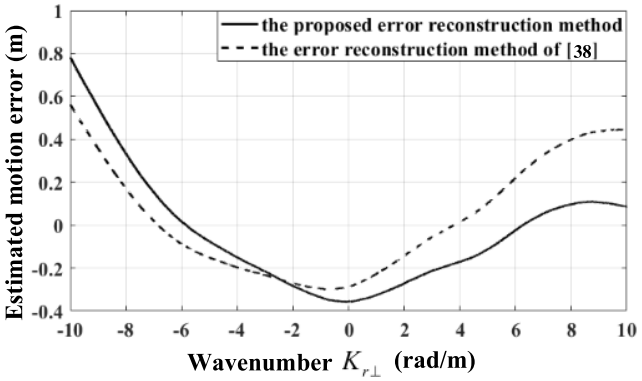


Fig. 16. Whole error function of three continuous full-apertures in the $K_{r\perp}$ domain.

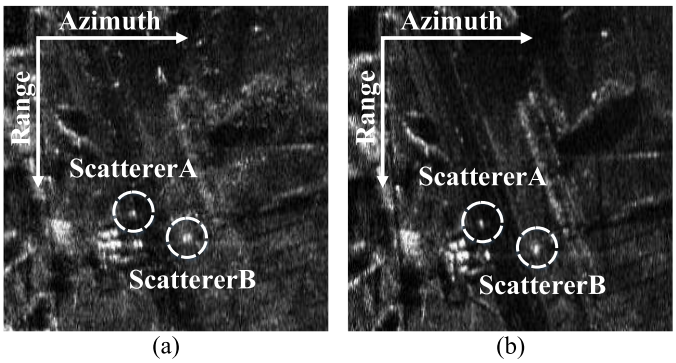


Fig. 17. Zoomed BiSAR images and prominent scatterers. (a) Zoomed BiSAR image of scatterer A and scatterer B obtained by the error reconstruction method of Zhang *et al.* [38]. (b) Zoomed BiSAR image of scatterer A and scatterer B obtained by the proposed error reconstruction method.

obtained and shown in Fig. 16. In Fig. 16, the dashed curve denotes the reconstructed phase error function obtained by the error reconstruction method of [38], and the MOCO result is shown in Fig. 15(c). It can be seen that the quality of the image is improved, and the entropy of the image decreases to 11.850. However, there is still some defocusing in the image. By utilizing the proposed error reconstruction method, the whole error function of three continuous full-apertures is obtained and described by the solid curve in Fig. 16, and the MOCO result is shown in Fig. 15(d). By comparing Fig. 15(d) with (c), it can be noted that the focusing quality is improved obviously, and the entropy of the image decreases to 11.613, which shows a higher performance of the proposed algorithm.

Fig. 17 presents the zoomed BiSAR images and prominent scatterers obtained by two kinds of error reconstruction methods. Comparing Fig. 17(a) and (b), scatterer A and scatterer B can be clearly distinguished in the BiSAR image obtained by the proposed algorithm. Then, the azimuth responses of two prominent scatterers are particularly evaluated. Fig. 18 shows azimuth responses of scatterer A and scatterer B obtained by two kinds of error reconstruction methods. From Fig. 18(a) and (b), it is seen that the proposed algorithm can achieve higher focusing quality. The obtained resolution of scatterer A in azimuth is about 0.31 m, and the resolution of scatterer B in azimuth is about 0.56 m. By utilizing the error reconstruction method of Zhang *et al.* [38], the obtained

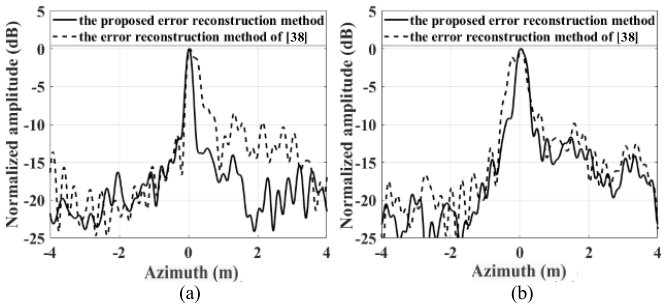


Fig. 18. Azimuth responses of prominent scatterers obtained by two kinds of error reconstruction methods. (a) Azimuth responses of scatterer A. (b) Azimuth responses of scatterer B.

TABLE III
FOCUSING PERFORMANCE EVALUATION FOR THE METHOD OF ZHANG *et al.* [38]

	Range direction		
	3dB width	PSLR	ISLR
Scatterer A	0.33m	-12.72dB	-8.84dB
Scatterer B	0.33m	-12.74dB	-8.73dB
	Azimuth direction		
	3dB width	PSLR	ISLR
Scatterer A	0.62m	-11.78dB	-8.64dB
Scatterer B	0.84m	-11.47dB	-8.35dB

TABLE IV
FOCUSING PERFORMANCE EVALUATION FOR THE PROPOSED METHOD

	3dB width	PSLR	ISLR
Scatterer A	0.33m	-12.82dB	-8.96dB
Scatterer B	0.32m	-12.79dB	-8.94dB
	Azimuth direction		
	3dB width	PSLR	ISLR
Scatterer A	0.31m	-12.83dB	-8.97dB
Scatterer B	0.52m	-12.28dB	-8.66dB

resolution of scatterer A in azimuth is about 0.62 m, and the resolution of scatterer B in azimuth is about 0.84 m. Then, the quantitative results in terms of 3-dB main lobe and ratios between main lobe and sidelobes (peak sidelobe ratio and integrated sidelobe ratio) are given in Tables III and IV. From the comparisons, it is seen that the improved focusing quality demonstrates the performance superiority of the proposed algorithm.

VI. CONCLUSION

In this article, a new FTDA framework for BiSAR continuous imaging is developed. The redundant BP operations of traditional FFBP are reduced by the new FTDA framework so that imaging efficiency is improved significantly. In addition, the new FTDA framework is also integrated with data-driven MOCO. By utilizing the proposed error reconstruction method, the phase error obtained from the phase history domain in different polar coordinate systems is transformed into the spatial domain for reconstruction to remove the discontinuity and defocusing from the final image. Both focusing quality and computational efficiency are evaluated by simulation and raw

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data experiments, and the experimental results demonstrate the superior performance of the proposed algorithm.

ACKNOWLEDGMENT

The authors would like to thank the anonymous reviewers for their valuable comments to improve the quality of this article.

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